

SOUTH WOOD COUNTY AP, WI, USA

WMO: 726452

Lat: 44.359N

Lon: 89.837W

Elev: 311

StdP: 97.65

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 04826

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.7	-21.2	-28.0	0.3	-22.8	-25.6	0.4	-20.6	9.6	-9.8	8.8	-9.1	2.2	290	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	32.0	22.6	30.1	21.4	28.5	20.7	24.1	29.3	23.1	27.9	22.1	26.8	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	18.0	26.7	21.6	16.8	25.7	20.8	16.0	25.0	74.1	29.5	70.1	27.9	66.3	26.5	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	8.1	7.3	DB	-28.3	34.6	2.7	2.0	-30.3	36.0	-31.9	37.2	-33.4	38.3	-35.4	39.8	
			WB	-28.5	25.8	2.7	1.3	-30.5	26.8	-32.1	27.6	-33.6	28.3	-35.6	29.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-8.4	-6.6	-0.1	7.3	13.9	19.1	21.7	20.3	16.2	8.9	1.7	-5.4
	DBStd	11.78	6.86	6.60	6.62	5.32	4.79	3.67	3.24	3.11	4.53	5.15	5.50	6.39
	HDD10.0	2303	571	464	320	113	16	0	0	0	5	84	252	478
	HDD18.3	4262	829	698	571	333	153	34	8	15	92	294	498	736
	CDD10.0	1381	0	0	8	31	137	274	364	321	192	51	4	0
	CDD18.3	298	0	0	1	1	16	58	113	77	29	3	0	0
	CDH23.3	2869	0	0	4	26	198	557	1117	667	269	31	0	0
	CDH26.7	850	0	0	0	4	54	152	389	180	68	3	0	0
Wind	WSAvg	3.3	3.4	3.5	3.5	3.9	3.6	3.1	2.9	2.7	2.9	3.3	3.3	3.3
Precipitation	PrecAvg	803	24	22	45	76	99	118	96	100	87	63	42	32
	PrecMax	1047	64	53	134	139	215	239	249	254	215	150	117	82
	PrecMin	620	7	7	4	29	36	43	20	14	15	7	1	6
	PrecStd	97	12	10	27	30	36	57	53	52	49	32	26	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	12.2	22.5	26.9	31.2	32.5	35.3	32.8	31.3	26.8	19.4	12.3
		MCWB	3.8	8.9	15.5	16.2	20.5	22.9	24.4	23.0	22.4	19.0	12.9	10.3
	2%	DB	4.6	7.3	17.2	22.6	27.8	30.1	32.4	30.7	28.6	23.4	15.4	7.4
		MCWB	2.2	4.2	11.2	13.3	18.2	21.2	23.1	22.0	20.9	17.1	11.3	4.8
	5%	DB	2.3	4.2	13.0	19.7	25.4	28.1	30.5	28.5	26.5	20.7	12.7	4.7
		MCWB	0.8	2.0	8.7	11.6	16.8	20.3	21.9	21.0	19.1	15.5	8.9	2.8
	10%	DB	0.9	2.3	9.5	17.1	22.7	26.8	28.7	27.1	24.1	17.6	10.0	2.6
		MCWB	0.0	0.5	5.5	9.7	15.2	19.3	20.8	20.0	18.1	13.6	7.0	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	8.0	15.9	17.4	22.1	24.6	26.1	25.1	23.4	21.0	14.4	10.7
		MCDB	7.0	11.5	21.6	23.5	28.7	30.1	32.3	29.9	29.1	25.0	17.7	11.8
	2%	WB	2.4	4.4	12.7	14.8	20.0	23.2	24.5	23.6	22.0	18.4	12.0	5.7
		MCDB	4.3	6.9	15.6	20.8	25.2	28.3	29.8	28.1	26.7	21.4	14.8	6.9
	5%	WB	1.0	2.3	9.0	12.6	18.2	21.9	23.4	22.4	20.8	16.2	9.3	3.0
		MCDB	2.0	4.4	13.4	18.1	23.2	26.5	28.4	26.6	24.2	19.9	12.1	4.6
	10%	WB	-0.3	0.6	5.7	10.3	16.7	20.5	22.4	21.3	19.4	13.9	7.1	1.2
		MCDB	0.9	2.0	9.2	15.9	21.1	24.6	27.0	25.3	23.2	17.4	9.6	2.4
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.5	10.3	11.8	12.3	11.8	12.2	11.6	12.1	10.6	8.9	7.8
		MCDBR	9.3	10.5	14.3	17.0	15.4	13.8	13.7	13.1	13.9	14.1	13.3	9.9
		MCWBR	7.6	7.8	8.9	9.2	7.3	6.5	5.7	5.7	6.8	8.3	9.2	8.0
	5% WB	MCDBR	8.7	9.9	13.3	14.5	13.1	12.1	11.8	11.0	11.4	12.3	12.2	8.6
		MCWBR	7.4	7.5	8.8	8.9	7.0	6.5	5.9	5.6	6.6	8.3	9.3	7.3
Clear-Sky Solar Irradiance	taub	0.254	0.285	0.320	0.354	0.389	0.394	0.385	0.388	0.358	0.321	0.289	0.262	
	taud	2.402	2.311	2.376	2.360	2.310	2.325	2.356	2.348	2.428	2.521	2.527	2.452	
	Ebn at Noon	887	907	917	908	882	876	879	864	866	860	839	841	
	Edh at Noon	73	97	105	116	126	125	120	116	99	79	66	63	
All-Sky Solar Radiation	RadAvg	1.71	2.82	3.93	4.53	5.41	5.94	6.10	5.21	4.14	2.63	1.68	1.25	
	RadStd	0.20	0.25	0.44	0.45	0.41	0.44	0.30	0.34	0.30	0.30	0.17	0.14	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (50 neighbors)		(m)								
		N/A	N/A	-2.20	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page